In 1915 the equation was simpler still, because the term $+\Lambda gab$, which Einstein added two years later (and which I discuss below) did not yet exist.[fn15] R_{ab} depends on Riemann's curvature, and together with $½Rg_{ab}$ represents the curvature of spacetime; T_{ab} stands for the energy of matter; G is the same constant that Newton found: the constant that determines the strength of the force of gravity.

That's it. A vision and an equation.

Mathematics or physics?

I would like to pause, before continuing with physics, to make a few observations about mathematics. Einstein was no great mathematician. He struggled with maths. He says this himself. In 1943 he replied in the following way to a nine-year-old child with the name of Barbara who wrote to him about her difficulties with the subject: 'Don't worry about experiencing difficulties with maths, I can assure you that my own problems are even more serious!'[2] It seems like a joke, but Einstein was not kidding. With mathematics, he needed help: he had it explained to him by patient fellow students and friends, such as Marcel Grossman. It was his intuition as a physicist that was prodigious.

During the last year in which he was completing the construction of his theory, Einstein found himself competing with David Hilbert, one of the greatest mathematicians of all time. Einstein had given a lecture, attended by Hilbert, in Göttingen. Hilbert immediately understood that Einstein was in the process of making a major discovery, grasped the idea and tried to overtake Einstein and be the first to write the correct equations of the new theory Einstein was slowly building. The sprint to the finish line between the two giants was a nail-biting affair, eventually decided by a matter of just a few days. Einstein, in Berlin, ended up giving a public lecture almost every week, each time presenting a different equation, anxious that Hilbert would not get to the solution before him. The equation was incorrect every time. Until, that is, by a hair's breadth – just marginally ahead of Hilbert – Einstein found the right one. He had won the race.

Hilbert, a gentleman, never questioned Einstein's victory, even though he was working on very similar equations at the time. In fact, he left a gentle and beautiful phrase which captures perfectly Einstein's difficult relationship with mathematics, and, perhaps, the difficult relationship which exists generally between the whole of physics and mathematics. The maths that was necessary to formulate the theory was geometry in four dimensions, and Hilbert writes:

> Any youngster on the streets of Göttingen[fn16] understands geometry in four dimensions better than Einstein. And yet, it was Einstein who completed the task.